ANSWER KEY

InquiryMind
LEARN · PLAY · WONDER

FACT CHECK · B2 · CYCLONES AND CLIMATE

Cyclone facts

★☆☆
Grade 4

Learning goals

- Order the stages of a cyclone.
- Label a cyclone.
- Recall safety facts.

1 Number the steps in the order a cyclone forms. [4]

5 The storm grows into a cyclone.

1 The Sun heats the sea above 27 °C.

4 Air rushes in and starts to spin.

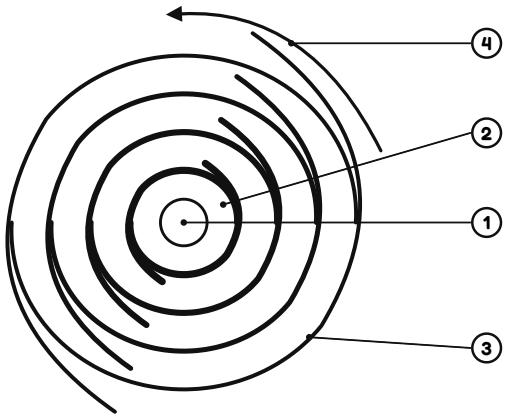
2 Water evaporates and warm, damp air rises.

3 Huge clouds form as the vapour condenses.

2 Label the cyclone. [4]

LABELS

rain bands **eye: calm, clear sky** **winds spin anticlockwise**
eye wall: strongest winds and rain



1 eye: calm, clear sky

2 eye wall: strongest winds and rain

3 rain bands

4 winds spin anticlockwise

3 Match the danger to what it does. [3]

Draw a line to match each pair, or write the letter.

strong wind ●	→ C	● **A** causes floods and landslides
heavy rain ●	→ A	● **B** pushes salty sea water onto the land
storm surge ●	→ B	● **C** blows down trees and houses

4 True or false? [4]

Statement	True	False	If false, correct it
1. Cyclones form over cold water.		✔	They form over warm sea water.
2. Signal 10 means great danger.	✔		
3. A cyclone shelter is built on tall pillars.	✔		
4. Burning fuels adds carbon dioxide to the air.	✔		

ANSWER KEY

InquiryMind
LEARN · PLAY · WONDER

THINK IT THROUGH · B2 · CYCLONES AND CLIMATE

Cyclones and climate

★★☆
Grade 4

Learning goals

- Explain cyclone formation.
- Link human actions to climate change.

1 **Why do most cyclones in the Bay of Bengal happen in April–May and October–November, and not in January?** [2]

The sea is warmest before and after the monsoon; in January the sea is cooler, so there is not enough warm, damp air to feed a cyclone.

2 **Why do cyclones get weaker after they move over land?** [2]

Over land there is no warm sea water to give them heat and moisture, so they lose energy.

3 **Explain how salty water from a storm surge can cause problems for farmers for many months.** [2]

Salt stays in the soil and ponds; rice and vegetables cannot grow in salty soil and fish ponds are spoiled until rain washes the salt away.

4 **Complete the chain to show how human actions can make cyclones more dangerous.** [3]

5	cyclones can become stronger
1	people burn coal, oil and gas
4	the sea gets warmer
2	more carbon dioxide in the air
3	heat is trapped and the Earth warms

5 **Suggest two ways people in Bangladesh can adapt to stronger cyclones.** [2]

build more cyclone shelters and strong houses; plant mangroves along the coast; better warnings; raise embankments.

ANSWER KEY

InquiryMind
LEARN · PLAY · WONDER

INVESTIGATE · B2 · CYCLONES AND CLIMATE

Track a cyclone

★★☆
Grade 4

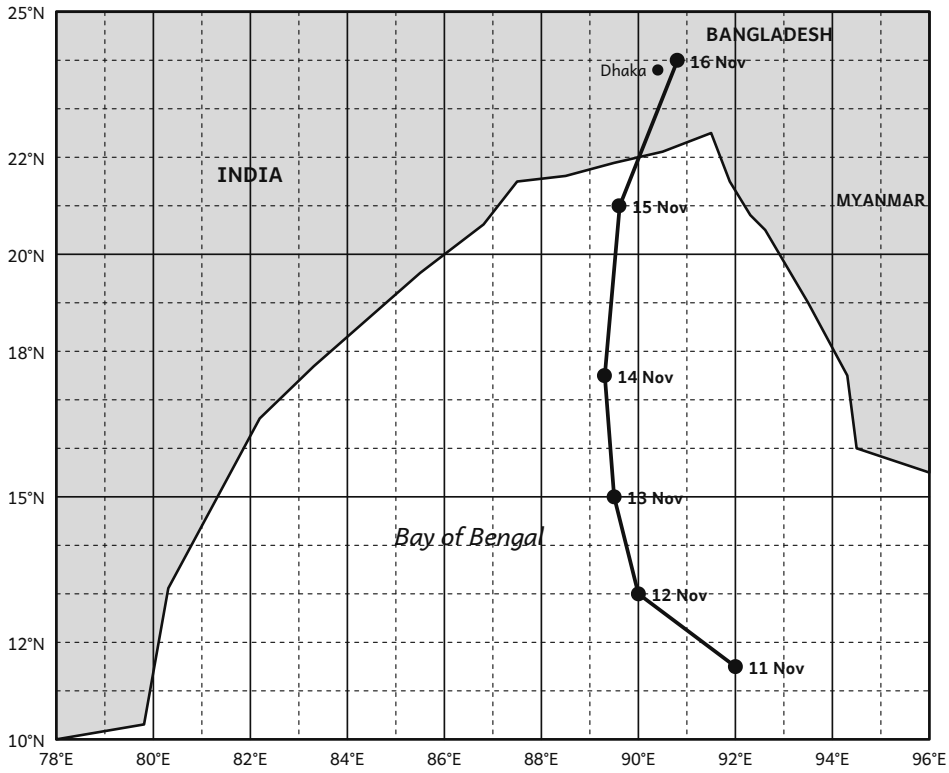
Learning goals

- Plot points using latitude and longitude.
- Describe a track.
- Use data to make predictions.

These are the positions of the centre of Cyclone **Sidr** in November 2007 (rounded). Plot each point on the map, join them in order, and write the date next to each point.

Date	11 Nov	12 Nov	13 Nov	14 Nov	15 Nov	16 Nov
Latitude (°N)	11.5	13	15	17.5	21	24
Longitude (°E)	92	90	89.5	89.3	89.6	90.8

1 **Plot the track.** [4]



2 **In which direction did Sidr mostly travel?** [1]

north (first north-west, then north, then north-east)

3 **On which date did it reach the coast of Bangladesh?** [1]

15 November (evening)

4 **About how far north did it move each day between 13 and 15 November?** [1]

about 3 degrees of latitude a day (15 → 17.5 → 21)

5 **Why is it important to track a cyclone several days before it reaches land?** [2]

So warnings can be given early and people have time to move to shelters with their animals and belongings.

ANSWER KEY

InquiryMind
LEARN · PLAY · WONDER

REAL-WORLD CASE STUDY · B2 · CYCLONES AND CLIMATE

Ready for the storm: a family cyclone plan

★★★
Grade 4

Learning goals

- Use real warnings.
- Plan to keep people and animals safe.
- Think about helping others.

Science · Citizenship · Health · Maths

Rahima lives with her family in a village near the coast in Khulna. One evening the radio says the signal has been raised from **4** to **7**, and a cyclone may reach the coast in two days. Volunteers from the **Cyclone Preparedness Programme** walk through the village with megaphones and raise warning flags.

Rahima's family must decide what to take to the shelter, what to do with their goats and chickens, and how to help their grandmother, who cannot walk far. Thanks to warnings, shelters and volunteers, the number of people who lose their lives in cyclones in Bangladesh has fallen enormously since 1970.

Based on the Cyclone Preparedness Programme (Bangladesh Red Crescent Society and the Government of Bangladesh).

1 **Sort the items: which should Rahima's family take to the shelter?** [6]

SORT THESE

dry food **drinking water** **important papers in a plastic bag** **heavy furniture** **medicines** **television**

Take	Leave or protect at home
drinking water	television
dry food	heavy furniture
medicines	
important papers in a plastic bag	

2 **The signal goes up to 9. What does that mean, and what should the family do?** [2]

Great danger: the cyclone is close; go to the cyclone shelter now.

3 **How could the family keep their goats and chickens safe?** [1]

Take them to the raised ground near the shelter (many shelters have space for animals) or untie them so they can escape.

4 **How can the family help Grandmother? Who else in the village might need help?** [2]

Leave early and carry her or use a van; ask volunteers to help. Others: babies, pregnant women, people who are ill or disabled.

5 **Design a "cyclone ready" checklist poster for families.** [2]

Cyclone ready!

e.g. listen to the radio; know the way to the shelter; pack water, food, medicine, torch, papers; help neighbours.

ANSWER KEY

InquiryMind
LEARN · PLAY · WONDER

REVIEW AND CHALLENGE · B2 · CYCLONES AND CLIMATE

Show what you know

★★★
Grade 4

Learning goals

- Use key vocabulary.
- Answer test-style questions.

1 Find the key words.

O	J	O	D	C	T	K	F	S	Y
S	H	E	L	T	E	R	S	I	K
U	C	C	P	M	Y	J	U	G	W
C	Y	C	L	O	N	E	R	N	I
E	M	Y	I	U	T	W	G	A	N
U	G	G	N	B	R	R	E	L	D
P	F	I	P	E	J	T	E	U	K
N	S	T	O	R	M	N	P	E	J
B	W	A	R	M	K	K	F	Y	H
S	T	T	R	A	C	K	D	E	W

cyclone eye shelter signal storm surge track warm wind

2 Where do cyclones form? [1]

- ○ over cold mountains
- ● **over warm seas**
- ○ in deserts
- ○ over ice

3 The calm centre of a cyclone is called the... [1]

- ● **eye**
- ○ tail
- ○ surge
- ○ track

4 Draw a simple diagram to show how warm, damp air rising from the sea forms storm clouds. [3]

Storm clouds forming

Sea with arrows of vapour rising, a tall cloud above, arrows showing air rushing in at the bottom.

5 **Challenge. Bangladesh is planting mangrove forests along the coast. Explain how this can protect people from cyclones and how it helps the climate.** [3]

Mangroves slow down the wind and the storm surge, so less water and wind reach villages. Trees also take in carbon dioxide, which helps slow climate change.